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1. INTRODUCTION

A. FOREWORD

In the realm of education, accurate and efficient monitoring of student attendance stands as a vital pillar for academic success. However , the conventional method of relying on manual class attendance lists has proven to be an Achilles' heel , shackling the educational institution with a myriad of weaknesses and drawbacks.

As such this report aims at unveiling the pitfalls of manual class attendance, bridging the gap using a biometric system authentication method in this case the fingerprint system as well defining system requirements obtained through a number of requirements gathering techniques.

B. ATTENDANCE WORK FLOW

As aforementioned, the conventional manual system although useful has a variety of loud shortcomings amongst which include;

- **Proneness To Human Errors**
  The reliance on manual class attendance list exposes the system to inherent fallibility of human involvement such as signing attendance in the wrong spot, errors at analytical level and even loss of attendance sheets which may lead to inaccurate data entry results.

- **Time Consuming And Inefficiency**
  The manual process of recording attendance is a laborious and time consuming task for lecturers. Valuable teaching time is consumed as lecturers meticulously call out and mark attendance diverting focus away from lectures. In cases where the list is passed round , it becomes time consuming and causes distraction from ongoing lectures. Lastly, final compilation and managing of multiple attendance sheets across different classes and courses becomes a logistical nightmare.

- **Vulnerability To Data Loss**
  Attendance records maintained in paper-based list are susceptible to loss, damage and unauthorized access.

- **Fraud**
  Fraudulent practices such as signing attendance of absent individuals becomes very likely in such a system.

These are but a few. As such fingerprint systems are proposed, needing a variety of requirement gathering techniques to enhance efficiency.

2. REQUIREMENT GATHERING AND TECHNIQUES.

In the realm of software development , where innovation and precision converge, the journey to success begins with an ardent quest to uncover the true essence of user needs and system requirement. Hence harnessing the power of requirements gathering techniques becomes paramount. In this respect , the following techniques are used;

- Brainstorming.
- Role play and Observations
- Interviews.

A. BRAINSTORMING AND RESEARCH

Through this technique, a vast scope of ideas converged and collapsed together gearing the requirement process towards identifying user roles, their requirements as well as prioritizing them. Research from online documentations and publications further solidified this.
As a result, three core users are defined here;

- Administrators
  These kind of user are involved in managing the system. They validate and thus activate lecturer's account as well perform system integration linking necessary data such as student and course data to the system.

- Lecturers
  These are the primary users involved with recording attendance, updating student status as well as accessing reports.

- Students
  They are indirectly impacted by the system as it affects their attendance records and performance evaluation. They validate their attendance using their fingerprints.

A number of other stakeholders come into play but are not deeply involved with the system. They are tertiary at most.

B. ROLE PLAYING AND OBSERVATIONS

By donning the personas of end users, stakeholders or system actors and transcending bias perspectives, immersion is achieved. Crowning this with meticulous attention to detail

gotten from observations, it becomes easy to clearly define functional and non functional requirements as well as any system or additional requirements

A real live scenario is observed such that authentication points are installed in each physical class. Students merely have to place their fingers on the biometric sensors. For this to be possible though, lectures need to have launched a session. This requires them to own a verified account. The login page of the application shows the option permitting lecturers to create an account. This is done using usernames, email as well as creating a password. After this, administrators confirm registration and associate lecturers to course as viewed form the database assuming that a course can have up to 3 instructors. They also upload all necessary data such as student list according to the lecturer's workload to the system. Once the account is activated, the lecturer can start using the system. This include setting up class sessions to permit attendance to be automatically populated into the system, recording and updating attendance and generating reports based on calculated statistics.

This begins unveiling aspects of requirements the system will need to perform in such a scenario.

C. INTERVIEWS

Through thoughtful questioning, active listening and empathetic engagement validating and verifying requirements as well as managing the changes is achieved.
Technical questions such as how the system will ensure transparency of attendance (eliminating false attendance checking), how additional updates during software utilization will be made as well as cases in which the lecturers pardon attendance or impose absence on students which will provide additional insight to the requirements.

For instance, installing biometric devises in each class taking into consideration that no two individuals have the same fingerprints ensures attendance transparency, using SDLC models such as the agile model, code documentation and version control software such as GitHub enables additional system update and patches, and given lecturers higher student priority permits absence pardoning or attendance imposing.

Below is a table denoting the interviews performed with some stakeholders, as well as their response and worries;

REQUIREMENTS GATHERING DONE BY INTERVIEWS FOR BIOMETRIC STUDENTS' ATTENDANCE MOBILE APP

NAME OF STAKEHOLDER INTERVIEWED	ROLE	QUESTIONS ASKED	ANSWERS
1) Dr Fremwen Alain	Lecturer	There will be features and functionalities like: -Real-time attendance tracking -Automated notifications -Reporting and attendance tracking So in your opinion, is there any desired functionalities that could be added in the app?	For me I think if a prototype could first be done, where I will use the app with all these functionalities and it is efficiency, then the real application could now be built. First do it and test if its working or if it could be implemented well.
2) Dr Mwebi Ekengoue Clautine	Administrator	How do you think this app can be good in a school milieu like FET?	I think if it reduces work for lecturers and its performance is good, then it'll have a positive impact on our school system.
3) Madam Becky (Works at the secretariat)	Administrator	What do you think about implementing such a technology in our faculty at various levels or departments to record students' attendance?	First thing we should think of before implementing such a system is if people will accept it. Like the culture and mindset of our present students. Are they going to take care of the device that will carry this app, will the university accept this new system and is it only feasible to really implement.
4) Dr Sop Deffo Lionel	Lecturer	• What's your take concerning the implementation of such an app in school system? • How would you like the mobile application to handle network connectivity issues?	• First, the lecturer should be the one to activate -maybe using his fingerprint, a code, or any other authentication measure- the attendance recording in the app for his/her course. Such if

		• What about the integration of the system to other school management systems? Is it necessary? • What do you think about privacy and security of the application with respect to users' data? • What of the customization of the app? Should different roles and access levels be given to lecturers, students' and administrators? Say for instance a lecturer wants to give like 5marks for attendance for his particular course, and another one 2marks and another 10marks. • How would the application handle different attendance scenarios such as late arrivals, early departure or absences?	he/she doesn't activate, no student should be able to mark his attendance. • I think the system should be always available with or without internet connection. Like it could even still be offline • Obviously integration might be necessary but I don't think it's feasible in our milieu. It might even be feasible but not possible because of the so many authorizations you need. • The notion of security is not necessary or really needed. Cuz in your database you can have just students' matricule to identify each student so they wouldn't even to know what data of them is stored in the app except their attendance. • I think that will be an additional feature if we want to do that. Cuz we want our

			system to take attendance and not process attendance. • Now that will depend on the lecturer's decision. Say for example if a student leaves before the end of the class and has not taken permission from the lecturer the student could be recorded absent. So the lecturer should have access and a user role where he/she can put a student absent or present?
5) Mr Patrick	Lecturer	• What's your take on the implementation of such an application?	• For us lecturers and in a faculty like FET, I think the app should be able to reduce stress and work for the lecturer. Like the app should even be able to bring out a list of students for a particular course, I should just click only one button and I'm able to input attendance for students. • I should be able to gather attendance for students, correctly, unbiased and efficient but

			without any stress or work. • The process should be collected, done and already documented with minimum stress for the lecturer. I want a method which is faster in taking attendance like the app but easy in recording. • I should be able to have pure and uncompromised data (data integrity), easy usage of app, then the attendance recorded could be easily converted into any format. Then that's the type of solution I need.

From the above table a number of requirements were obtained most of which were user related in nature while others played a fundamental role in predicting a 'current' ideal performance scenario. As such, a number of functional and non-functional system needs were also obtained

These 3 techniques give an insight into the system requirements (functional and non-functional) as well as user and additional requirements useful for ideal target system design.

The proceeding chapter dives into these requirements proper and what they imply.

3. REQUIREMENT SPECIFICATION

Harnessing the power of the above requirement gathering techniques, an insight into certain aspects of the application's functionalities is derived. The main class groups being the system requirements (Functional and Non-Functional) and User requirement and others. For convenience, they are classified into the following ;

A. FUNCTIONAL REQUIREMENTS

These are typically expressed as features, tasks or actions the system should be able to carry out. They specify the behavior and functionalities of the system.

- **Login and Registration**
  Permitted users should be able to create accounts and log into their dashboards after verification. From there they should be able to use the accounts whether for recording or management.

- **User Management**
  Application should permit loading of user info such as student list, assigning roles such that lecturers can modify final list; as well as deleting user accounts.

- **Attendance Recording and Management**
  System should be capable of biometric data capture (fingerprint capture in this case) , recording attendance, account for attendance as well manage various attendance types (present, absent, excused) and even allowing manual attendance entry in exceptions or emergencies.

- **Data Management**
  The software should be able to store attendance securely, generate reports and export data. It should also be able to integrate with existing school information system and provide real time data capture and historical data records.

B. NON-FUNCTIONAL REQUIREMENTS

These specify the qualities and constraints the system should possess. Here the focus is on attributes which permit the system to perform the tasks for which it was created

- **Security**
  The application should ensure confidentiality and integrity of biometric data, employing encryption and security storage mechanisms. For instance, the use of passwords and need to login enhances security.

- **Usability And User Experience**
  The application should have an intuitive and user- friendly interface. It shouldn't be complicated to use and colors and physical feels should be well balanced.

- **Performance**
  The application should have a minimal response time, handle concurrency in usage automatic online responding. The specification biometric (fingerprint) data capture her is set to a maximum of 5 seconds.

- **Scalability**
  The application should be able to handle increasing numbers of users without significant degradation in performance. Cloud storage can be used to support the ever-growing information as the system scales.

- **Platform Compatibility**
  The application should be compatible with major mobile platforms such as Android and iOS. Consequently, a cross platform development framework, React-Native is used.

- **Data Privacy**
  The application should comply with relevant privacy regulations for instance GDPR; and implement measures to protect user data. Administrators should see to this.

C. USER AND ADDITIONAL RQUIREMENTS

- To ensure ability for updates and patches, this application makes use of the Agile SDLC model, code documentation and version controls such as Git and GitHub.

- The system should meet the requirements of various core user such as admins and lecturers needs by providing reporting features to track user activities. For instance track the percentage attendance of students and generate repots.

- The application should meet compatibility requirements with existing system as well as cross platform compatibility.

- The application should meet integration requirements to permit required data loading.

- The application should ensure computational accuracy especially during report generation

To conclude, it is observed that through the aforementioned requirement gathering techniques (Brainstorming, Role Playing and observations; and interviews), system requirements are defined. These requirement portray the advantage of the biometric fingerprint system over the manual one and thus backing their necessity.

4. REFERENCES

Development of Fingerprint Biometric Attendance system pdf

Implementing Student Attendance System Using Fingerprint Biometrics.pdf for Kolej University poly-tech

Reviews and worries obtained from stake holders (students, lecturers and bureaucratic personnel(admins))

Researches from Google and Wikipedia